

Practical-1

Name: Ayush Borkar

PRN:- 202501050003

1.1.1 Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula: where: is the mass of the object (in kilograms). is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Code:

```
mass = float(input()) velocity
```

```
= float(input())
```

```
momentum = mass*velocity print(f"{momentum:.2f}kgm/s")
```

The screenshot displays a web-based IDE interface. At the top, the browser address bar shows the URL: `mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5f/698c54860aba96214c8b7d60/6773e9c2...`. The IDE header includes the 'CDETANTRA' logo, navigation links, and a user profile section with the email '202501050003@mitaoe.ac.in' and a 'Logout' button. The main workspace shows the execution results of a program. A summary bar at the top indicates '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. Below this, two test cases are detailed. Test case 1 shows an expected output of '5.0', '10.0', and '50.00kgm/s' matching the actual output. Test case 2 shows an expected output of '2.3', '4.8', and '11.04kgm/s' matching the actual output. The bottom of the interface features a navigation bar with buttons for '< Prev', 'Reset', 'Submit', and 'Next >'.

1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer as input. Depending on the number of digits in .

Constraints:

$$1 \leq \leq 999$$

Input Format:

- The input consists of a single integer .

Output Format:

- If is a single-digit number, print its square.
- If is a two-digit number, print its square root (rounded to two decimal places).
- If is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Code:

```
n=int(input()) if
0<=n and n<=9:
    print(n*n)      elif
10<=n and n<=99:
    print(f"{n**(1/2):.2f}") elif
100<=n and n<=999:

    print(f"{n**(1/3):.2f}") else:
    print("Invalid")
```

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Explorer

Average time: 0.004 s (4.43 ms) Maximum time: 0.005 s (5.00 ms)

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 (5 ms) Debug

Expected output: 9
Actual output: 9

Test case 2 (4 ms) Debug

Expected output: 166
Actual output: 166

Test case 3 (4 ms) Debug

Expected output: 24
Actual output: 24

Test case 4 (5 ms) Debug

Expected output: 3456
Actual output: 3456

Invalid Invalid

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format YYYY-MM-DD
- The second line contains the second date in the format YYYY-MM-DD
- An integer representing the number of days between the given dates.

Code: from datetime import datetime

date1=input().strip() date2=input().strip()

d1=datetime.strptime(date1,"%Y-%m-%d")

d2=datetime.strptime(date2,"%Y-%m-%d")

difference = d2-d1 print(difference.days)

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Explorer

Average time 0.007 s 7.00 ms Maximum time 0.008 s 8.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8 ms Debug

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2 6 ms Debug

Expected output

2014-07-02

2014-07-11

9

Actual output

2014-07-02

2014-07-11

9

Debugger

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Code:

```
num = int(input()) reversed_num=  
int(str(num)[::-1])  
print(reversed_num)
```

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Explorer

Average time 0.005 s 5.40 ms Maximum time 0.008 s 8.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output 5367 7635

Actual output 5367 7635

Test case 2 8 ms Debug

Expected output 8132 231

Actual output 8132 231

Debugger

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format: **<number> x**

<multiplier> = <result> Code:

```
num = int(input()) for
```

```
i in range(1,11):
```

```
    print(f"{num} x {i} = {num * i}")
```

The screenshot shows the CodeTANTRA web interface. At the top, there's a navigation bar with 'Course', '2304102 EDS: Activity No. 1', and 'stylecentralog-del/EDS-Lab-As'. The main header displays 'CODETANTRA' and 'Home Learn Anywhere'. The right side of the header shows '202501050003@mitaoe.ac.in', 'Support', and a 'Logout' button. The left sidebar contains 'Explorer' and 'Debugger' tabs. The main content area shows two test cases. Test case 1 has an expected output of 8 and an actual output of 8. Test case 2 has an expected output of 5 and an actual output of 5. The interface also displays performance metrics like 'Average time' and 'Maximum time'.

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer , the number of courses.
- The second input will be integers representing the marks of the student in each of the courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Code:

```
def cal(marks, total_courses):  
    if any(mark < 40 for mark in marks):  
        return "Fail"  
    total_marks = sum(marks)  
    aggregate_percentage=(total_marks/(total_courses *100))*100  
    if aggregate_percentage >75: grade="Distinction"  
    elif aggregate_percentage >=60: grade= "First Division"  
    elif aggregate_percentage >=50: grade= "Second Division"  
    elif aggregate_percentage >=40: grade= "Third Division"  
    return (aggregate_percentage,grade)  
num_courses= int(input())  
marks= list(map(int,input().split()))  
result= cal(marks,num_courses)  
if result == "Fail":  
    print("Fail")  
else:  
    aggregate_percentage,grade=result  
    print(f"Aggregate Percentage: {aggregate_percentage:.2f}")  
    print(f"Grade: {grade}")
```

The screenshot displays a web browser window with the URL `mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5f/698c54860aba96214c8b7d61/67731211f...`. The page header includes the logo for "CDETANTRA" and navigation links. The main content area shows the execution results of a Python program. At the top, a summary bar indicates "5 out of 5 shown test case(s) passed" and "5 out of 5 hidden test case(s) passed". Below this, five test cases are listed, each with a green checkmark and a "Debug" button. The first two test cases are expanded, showing "Expected output" and "Actual output" columns. For Test case 1, the expected output is "4" and the actual output is "4". For Test case 2, the expected output is "5" and the actual output is "5". The third, fourth, and fifth test cases are collapsed. At the bottom of the page, there are buttons for "< Prev", "Reset", "Submit", and "Next >".

1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first terms of the Fibonacci series.

Input Format:

- A single integer representing the number of terms to generate.

Output Format:

- A single line containing the first terms of the Fibonacci sequence, separated by spaces.

Code: def

fibonacci(n):

if n == 1:

return 0

elif n == 2:

return 1

else:

return fibonacci(n - 1) + fibonacci(n - 2)

n = int(input())

for i in range(1, n + 1):
print(fibonacci(i), end=" ")

The screenshot displays the CodeTANTRA web interface. At the top, the browser address bar shows the URL: `mitaoe.codetantra.com/secure/course.jsp?euid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5f/698c54860aba96214c8b7d61/6682597f...`. The CodeTANTRA logo and navigation links (Home, Learn Anywhere) are visible. The user's email `202501050003@mitaoe.ac.in` and a `Logout` button are in the top right. The main content area shows the execution results of a Python program. It includes a table with performance metrics: Average time (0.019 s, 18.75 ms) and Maximum time (0.054 s, 54.00 ms). Below this, it states '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. Two test cases are detailed: Test case 1 with expected output '0 1 1 2 3' and actual output '0 1 1 2 3'; and Test case 2 with expected output '0 1 1 2 3 5 8 13 21 34 55 89 144 233 377' and actual output '0 1 1 2 3 5 8 13 21 34 55 89 144 233 377'. The interface also features a left sidebar with 'Explorer' and 'Debugger' tabs, and a bottom navigation bar with buttons for '< Prev', 'Reset', 'Submit', and 'Next >'.

1.2.3. Pattern -1

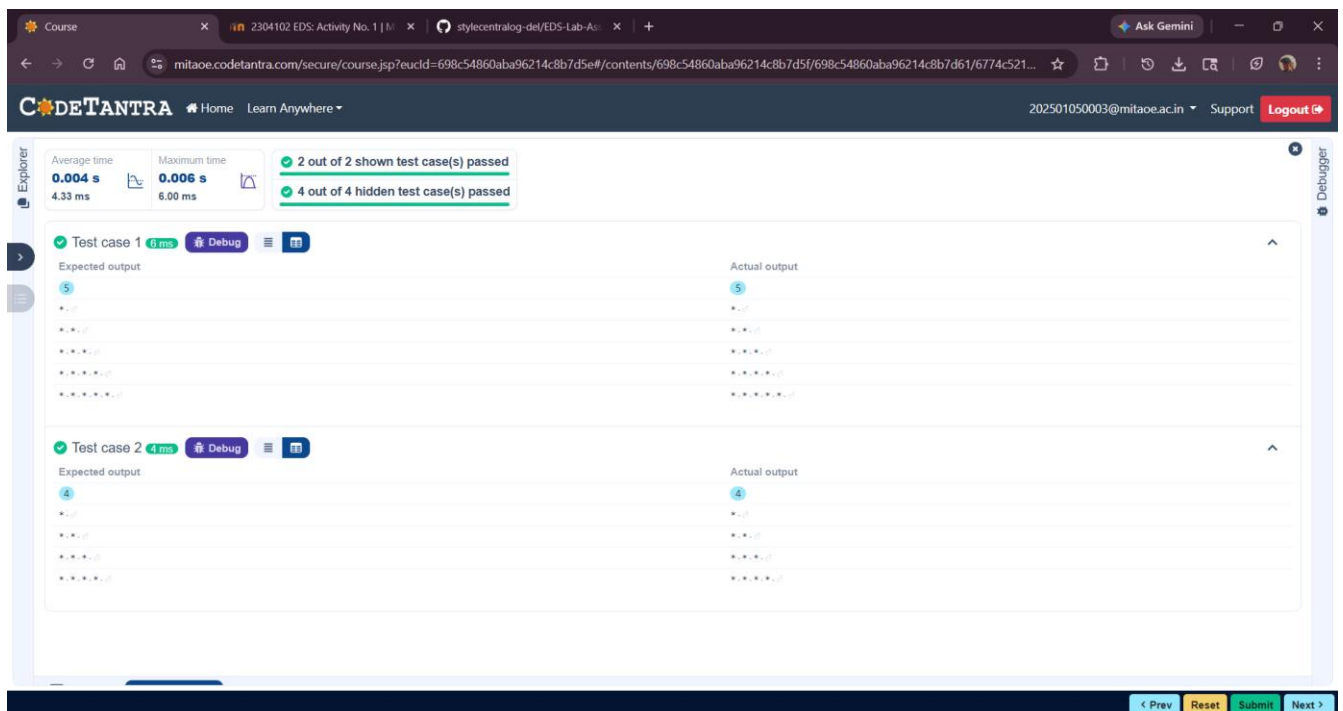
Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern. **Output Format**
- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Code:

```
n = int(input())
for i in range(1, n+1):
    print('* ' * i)
```



1.2.4. Pattern -2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Code:

```
n = int(input())
for i in range(1, n+1):
    for j in range(1, i+1):
```

print(j,end=" ")

print()

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Explorer

Average time
0.005 s
6.00 ms

Maximum time
0.006 s
6.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output

5

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Actual output

5

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Test case 2 5 ms Debug

Expected output

8

1 2

1 2 3

1 2 3 4

1 2 3 4 5

1 2 3 4 5 6

1 2 3 4 5 6 7

1 2 3 4 5 6 7 8

Actual output

8

1 2

1 2 3

1 2 3 4

1 2 3 4 5

1 2 3 4 5 6

1 2 3 4 5 6 7

1 2 3 4 5 6 7 8

< Prev

Reset

Submit

Next >

Debugger